

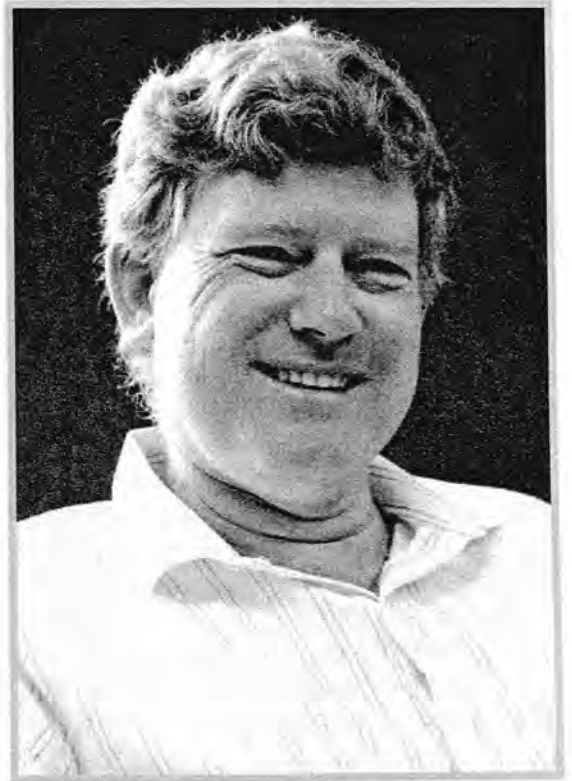
SETI and Human Behavior: Human Responses to an ETI Signal Detection

*Members: Michael Ashkenazi (coordinator), Robert Arnold,
Gary Coulter, Langdon Gilkey, Roger W. Heyns, Alex Inkeles,
Michael J. Klein, Vivian Sobchack*

Introduction

The human behaviors that would manifest themselves if an ETI signal were detected can be examined by using approaches from an array of disciplines – the “behavioral sciences.” The behavioral sciences are unique among the sciences in two ways. First, they deal with subjects – people – who respond to phenomena in unpredictable and ambivalent ways. Second, most behavioral study is done at a second remove: it is usually neither possible nor desirable to manipulate human subjects under controlled conditions. In the context of our chapter, these constraints imply that any statements about likely human responses to various detection scenarios must be preliminary.

Although there are few data on how humans would react to a scientific announcement of detection of a signal of ETI origin, we do have limited information on responses of some selected populations to the idea of the existence of ETI's and information about how people react to events that are somewhat analogous to SETI research and an ETI detection. From these we can postulate some possible human responses to detection of an ETI signal.



Michael Ashkenazi, of the Ben Gurion University.

To begin, we attempt to identify the factors that would initiate, influence, and channel individual and group responses to the detection of an ETI signal. These considerations provide a foundation for (eventual) predictions

of the behaviors themselves. A brief overview of available data on modern public acceptance of the existence of ETI's is a logical starting point.

Attitudes About the Possible Existence of ETI

Compared with voluminous literature compiled by scholars and philosophers describing historic and modern views of ETI, the scientific analysis of those views is in its infancy. There is, however, a small body of survey literature on contemporary opinions on the subject. These data have been acquired from small samples, limited populations, and (almost exclusively) Americans in the US. Two main issues have been explored in these studies: belief in the existence of ETI, and views of the desirability of contact with ETI.

Belief in the Existence of ETI

Recent surveys show that a considerable number of respondents accept the existence of ETI's as a fact. Only a small minority is sure that ETI's don't exist. Lavrakas and Rosenbaum (1978) constructed a scale based on 24 questions related to belief in the existence of ETI. On the scale, higher scores show stronger overall belief in the existence of ET life. The mean score on this "BEXTL" (Belief in Extra Terrestrial Life) scale for a sample of 378 respondents was 36.17 out of a possible 48. In another study, Bainbridge (1983) found that about 20% of his respondents agreed with the statement, "Intelligent life probably does not exist on any planet but our own." An important historical source for understanding views of ETI is the Gallup Poll. Some of the Gallup surveys have asked Americans their opinions about the existence

of ETI since 1964 (Pinotti, 1988). The responses show that the percent of Americans who believe that ETI's exist is significant and has increased over the past two decades (Table 3-1). An unpublished survey by Ashkenazi (1991) showed that 66% of Israeli students questioned (n=139) believe ETI's exist and that only 8% are certain that they do not.

Although the data are few, they (and anecdotal evidence) support the belief that acceptance of the existence of ETI's is widespread, at least in North America.

Belief in the Desirability of Communication

Various studies of popular views of ETI have taken different disciplinary slants and have used different criteria to formulate survey categories. Few have distinguished between mere signal reception and the more inclusive and dramatic idea of "contact". The latter encompasses deliberate efforts at one- or two-way communication (and visits via space travel), as well as passive signal interception.

Keeping in mind the conflation of "message," "signal," and "contact" in popular understanding, it is notable that a significant number of respondents thought that "contact" would be beneficial (or at least not harmful). Lavrakas and Rosenbaum (1978) concluded from their study that persons with a strong belief in ETI are also likely to believe that physical visits are probable within the next century. In Bainbridge's study (1983), about 42% of the respondents agreed that humanity should attempt communication with ETI. Most of the remaining respondents answered "don't know" to this suggestion and a minority felt that communication should not be attempted. Thus, acceptance of the existence of ETI's

Year	Believe ETI exist (% of those surveyed)	Skeptical or deny ETI (% of those surveyed)	Unsure (% of those surveyed)
1987	50	34	16
1978	51	33	16
1973	46	38	16
1966	34	46	20

Table 3-1. Gallup Poll results for selected years (Pinotti, 1988).

seems correlated with beliefs that contact is desirable and visits from ETI's are likely.

The issue of desirability of communication with ETI is related to the issue of the consequences of communication. The data on this are very limited. Only one study, in which the authors examine some consequences of simple signal detection, addresses that question specifically (D. Tarter, and Peacock, 1990). Journalists and media experts questioned by Tarter and Peacock feel that signal detection would lead to confusion and excitement, but with little panic or hysteria.

Causes for Accepting or Rejecting ETI

Why do some individuals accept and others reject the idea that intelligent life exists elsewhere in the galaxy? Surveys find no strong causal links between basic beliefs and factors such as age or sex. More women adopt an "uncertain" stance toward the existence or non-existence of ETI's than do men (Bainbridge, 1983; Lavrakas and Rosenbaum, 1978). Older persons are more likely to be "uncertain" than younger persons. However

these correlations are too weak to have predictive value.

Greater confidence can be attached to relationships between education, religion, and the public's beliefs in ETI. Higher educational level and membership in some professions seem to correlate with greater interest in ETI. Bainbridge (1983) points out that professionals in two fields – astronomy and anthropology – exhibit unusually high interest in ETI and acceptance of ETI existence. He attributes this to the fact that professionals from both disciplines are engaged in a study of the universe as a whole and of "strangeness" in general, and are therefore preconditioned to accept the existence of ETI. Membership in most other professions is less strongly associated with belief in ETI, with one exception – there is a strong rejection of the existence of ETI's by social workers. This may be a result of the focus by social workers on the solution of immediate human problems, as Bainbridge notes. However, a similar inverse correlation was not found among psychology professionals, who might also be expected to be oriented

toward solution of immediate proximal problems.

No study has been able to fully address the question of how religion influences belief in ETI. Negative correlations were demonstrated between the acceptance of ETI and membership in fundamental Christian churches by Rosenbaum, Maier, and Lavrakas (1979). They found that 76% of the ministers of fundamentalist Christian churches (defined by them as "accepting all statements in the Bible as factually accurate") felt that their church position would be "ET life does not exist." Bainbridge's (1983) findings also support the idea that fundamentalist Christians tend to deny the existence of ETI. Bainbridge suggests that the reason may well be a "...generalized rejection of strangers..." in some Protestant churches that manifests itself in this case as disbelief in the very existence of ETI's. Rosenbaum and colleagues feel that the existence of ETI's is viewed as a threat by members of these churches. The cognitive dissonance that the possibility of ETI existence poses is resolved by denying any likelihood for that existence.

Summary of Preliminary Findings

The limited studies conducted to date suggest three important elements of belief in ETI for North Americans. First, a significant and (apparently) growing number of people in the populations studied do indeed believe that extraterrestrial intelligent species exist. Second, most of those surveyed felt that communication with such beings would be beneficial (or at least non-threatening) to humankind. Third, a number of underlying factors have been cited as possible reasons for acceptance or rejection of the idea of the existence of (or communication with) ETI. None

of these studies unambiguously point to the ways in which people will actually react if a signal is received. We can anticipate with some confidence that people who accept the existence of ETI as a matter of fact will be less likely to react negatively upon news of a signal receipt than those who deny that possibility. This particular hypothesis needs further and broader testing, however.

Human beings respond to events not merely as empirical phenomena but also as *meaningful* phenomena that are subject to endless personal and cultural interpretations. Human responses to "meaningful" events are usually multi-leveled (based on material need, historical experience, fantasy and desire, for example) and ambivalent (conscious and unconscious, logical and emotional). These factors insure that the same individual may respond differently even to the same simple stimulus at different times. As noted, it is rarely possible to predict, on the basis of any complex stimulus, precisely what a given individual, group, or culture will do. In the SETI context, this means that even a clear and detailed exposition or understanding of the circumstances surrounding the receipt of an ETI signal will not necessarily cause the newly-informed populace to react in predictable or scientifically "rational" ways. Given the social, cultural, economic, emotional, and experiential differences between individuals, even equal possession of the full range of information about a detected signal will not insure that all would react to it in the same ways. Responses to an ETI signal will make sense from the standpoint of the individuals involved, but they will not necessarily be uniform even for fairly homogeneous populations.

As specifically applied to the question of human responses to an ETI detection, the behavioral sciences are hindered by several obstacles. These are the absence of an analyt-

ic framework for assessing the variables that influence human behavior in this context, the complexity and variety of theories that purport to explain human behavior in general, the variability of individual responses to the “same” stimuli, and the absence of any previous exposure to an authentic ETI signal with which valid comparisons could be made. Historical analogs to events similar to a detection, as noted in an earlier chapter, are of limited value, and no methodological or theoretical framework has been attempted before this.

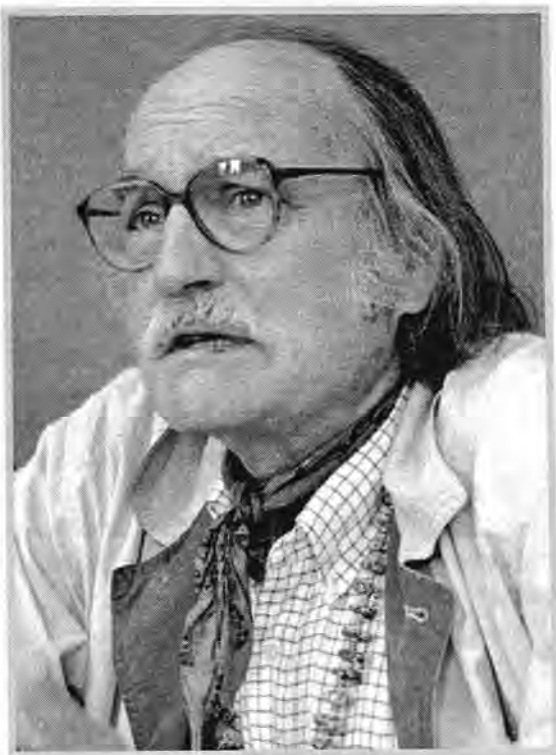
To address our topic, we have therefore drafted an analytical framework and methodology, adapted for this purpose, that borrows from the relevant behavioral sciences.

Anticipating Human Responses to the Detection of ETI

Attributes of People and Groups

We introduce a classification scheme for human data that is based on two distinct sets of traits. Drawing from anthropology, political science, psychology, and sociology, we can characterize every person at an individual level by a set of commonly used *social attributes*. At the collective level, groups and categories of individuals can likewise be characterized by a set of *social contexts*. These are properties of their environments that define them to themselves and in relation to others. Individual and group reactions to an ETI signal are traceable to attitudes and experiences that stem from (and are influenced by) these social indicators.

Certain attributes of individuals are more socially meaningful than others in any given culture. These attributes are partly inherent, partly self assumed, and partly defined by others, with overlaps that depend upon circumstance and context. Any particular attribute may be salient for a particular culture at a particular moment. For example, the attribute “eye color” is not as socially significant in the United States of the late 20th century as is the attribute “skin color.” Responses correlated with attributes are expressed both singly and in combination. For example, for various reasons and under different circumstances, the same individual may respond as a Black woman member of an intellectual elite, or as a Black woman, or as a member of the Black elite, or as a member of the elite, or as a woman, or in other ways.



Langdon Gilkey, of the religious studies departments of the University of Virginia and Georgetown University.

The attributes we identify here are salient in public life and likely to be relevant to SETI. These are an individual's gender, religion, "affiliative group," level of education, regional identity, access to power, and access to information.

Gender

Gender is a primary identifier of individuals in all societies. Based on sexual characteristics (but not completely dependent on them), the gender attribute usually goes beyond issues of biology. Every society has defined "female" and "male" roles, and has understandings about permissible and transgressive degrees of cross-over between roles and who may play the roles. In all societies, gender roles and behaviors are based on natural biological differences between women and men. However, subdivision by gender usually includes role expectations that result from a complex interplay between biology and social factors, as well. What is manifestly "male" behavior in one society may be seen as "female" in another. Men and women in some cultures may "choose" their gender to some degree, regardless of their sex. We have seen in the ETI context (Bainbridge, 1983) that women and men tend to react differently to questions about belief in ETI. Similar differences were found in a study by Ashkenazi (1991). If an ETI signal is identified, there may well be differences in the responses of the two sexes traceable to their different roles. Of course, responses due to gender will be intricately linked with responses stemming from other attributes of individuals, and other factors.

Religion

All societies have religions. This simple truism is complicated by the varieties of religious experience, adherence, and belief found in every society. All complex societies have several prominent religious alternatives, and religious affiliation (or non-affiliation) is an attribute of every individual. "Religiosity" includes a number of significant variables, not all of them material here. Two that are relevant are church affiliation and "world view", both of which significantly affect behavior (Yinger, 1970; Stark and Glock, 1969).

Membership in a religious organization or church affects individual behaviors (Durkheim 1947). The more an individual attends church activities, the more likely she or he is to behave like the other members and to espouse that church's religious views. In practice, each individual's understanding of the tenets of his or her religion differs from any other individual's, but the religious posture of each individual usually falls within the limits of the positions officially espoused by the church.

Religiosity will influence the degree to which an individual is willing to accept evidence of ETI life elsewhere. It will also influence the threshold of information that would be necessary to motivate that individual to react publicly to an announcement of that evidence. Different religions will inevitably view the advent of an ETI signal in different ways. Some will tend to view the possibilities implicit in signal reception as supportive of (or immaterial to) their own beliefs (Ashkenazi 1992), others might see the reception or possible reception of an ETI signal as threatening (Ashkenazi, 1991; Lavrakas, 1979). The degree of intensity with which each individual responds and whether the information about the signal will be accepted,

rejected, or transformed, will certainly be influenced by his or her religious views and affiliations as well as by other factors.

Affiliative Groups

"Race", "Ethnicity", "Place of origin", "Language group", "Caste", "Nationality" – these define social memberships for every person. These descriptors stem from inherent and regional characteristics of people and are often loosely defined. "Race," for example, is a widely-recognized yet partially-subjective construct that in fact has only a few quantifiable parameters (Montagu, 1963; van den Bergh, 1981). All individuals are affiliated with (or forced to affiliate with) a "race" whose characteristics are vaguely delimited in some instances and tightly defined in others (differing, for example, between the US and South Africa). "Ethnicity" is similarly a complex social assignment in which self selection and determinations by ethnic group members and by members of the society at large all play a part (Rex and Mason, 1968). The attribute "affiliative group" is used here to label all such social categories that are ascribed to each individual by the public (and also assumed by each individual as part of his or her public persona).

An affiliative group often defines roles for its member individuals. It is also used by others as an external gauge of the way a person "will" or "should" behave and indeed think. In other words, society at large often expects that all members of a particular ethnic group will think, feel, and behave in similar ways in some circumstances (Moynihan and Glazer, 1963). Identification with an affiliative group by individuals themselves, even when voluntary and self-selected, often limits each member's own expectations regarding his or her

behavior patterns and responses in various situations.

The significance of affiliative attributes for SETI is not easily specifiable. There are many affiliative groups within most cultures, and most individuals are members of more than one group. The implications of membership and behavioral expectations for a particular behavioral group vary from one context and society to another. Depending upon the circumstances and issues, identification with one affiliative group may momentarily overshadow affiliation with all others. For example, being Italian-American may overshadow other affiliative attributes for individuals involved in debates about Columbus and quincennial celebrations of his arrival in America. Being "white" may override being Italian-American in another circumstance, and being Italian-American in the United States is different from being Italian-American in Italy. It is certain, however, that efforts to mobilize any particular affiliative group in relation to the continued search for ETI signals will strongly influence the members of the affiliative group to respond in the way advocated.

Education

Formal education is prevalent in virtually every modern human society, but the mode, extent, and quality of education vary widely. In cultures with formal educational institutions, differences in educational level (roughly, number of years of formal schooling) and educational type (roughly, concentration on technical, scientific, humanistic, artistic, or professional training) can result in differences in behavior and receptivity to new ideas and circumstances. Based on the relatively few studies on the subject mentioned here, more extensive education will generally result in a

more positive response to SETI and to detection of an ETI signal (Bainbridge, 1983; Lavrakas and Rosenbaum, 1978; D. Tarter, 1989, 1991).

Incorporating information about SETI and ETI in educational programs and information media would ensure that detection of an ETI signal would be surprising to fewer people and that it might even be viewed by them as an expected occurrence. We may also hypothesize that the more the public is educated to recognize the possibility of the existence of ETI in the context of the size of the galaxy, the less the absence of a signal would appear to be a problem for SETI researchers. Ways of educating publics to recognize the possibility of detection of an ETI signal are discussed in chapter 5; the justification of such an effort lies in its ability to better inform the judgments from which normal patterns of human behavior originate.

Regional Identity

Regional identity is an attribute that links each individual with a geographic area: its landscape, history, and concerns. Geography provides a context within which people define and orient themselves. Differences in "region" subsume both inequities between regions and differences between and within nation states. Inequities between regions and differences between nation states create differing opportunities for participation for individuals in different regions, just as regional similarities create popular commonalities.

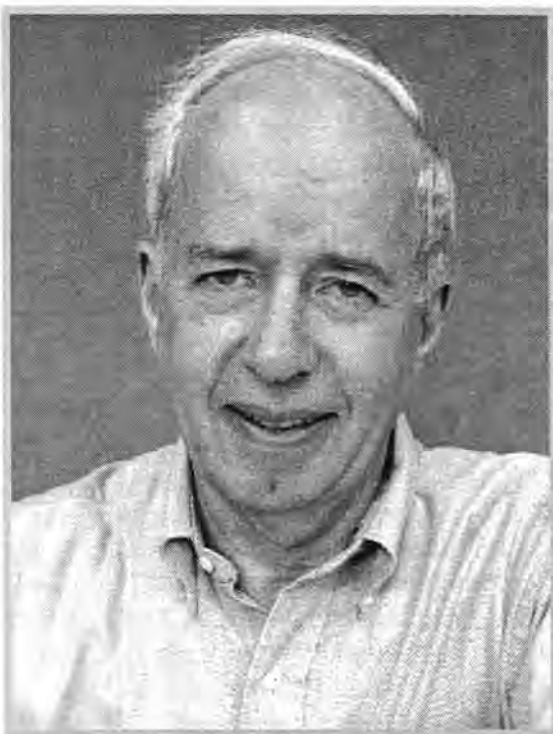
While people may feel united in aspects of regional identity at the national level, individuals may also differ in the degree of their regional allegiance. Examples are provided by differences between urban and rural perspectives and behaviors. As a result of emi-

gration, resettlement, and diasporic movements, individuals and groups may also be torn between allegiance to their adopted country of residence and perceived conflicting obligations to a previous "homeland". We can expect that pride and interest stemming from regional identity will affect many individuals' reactions to an ETI detection. In practice this may mean that the internationalist perspective adopted in this book and by the SETI enterprise generally may come into conflict with national and even regional sentiments. It is not unlikely that some Americans, for instance, would be reluctant to share information received from an ETI source through American-funded instruments with other countries or international bodies.

Access to Power

Individuals in all societies have differential access to influence and power. Powerful individuals respond to important events which they perceive as affecting them and upon which they can wield their influence. Such individuals will probably respond to news of an ETI signal to the degree that it is perceived as relevant to their interests. Individuals with greater access to power would be more inclined to actively respond to news of an ETI detection than would those with less access to power, even given equal information and comparability in other factors.

In many countries, established socio-political elites (often those with long-standing economic and political power) subtly determine many aspects of national mood and action. This is true, for instance, in Britain (Cohen, 1981) and in the US (Tocqueville, 1964; Verba and Kelman, 1987). This influence is not negligible. Elites with their own interests and agendas will probably affect the reactions of whole



Alex Inkeles, Senior Fellow at Stanford's Hoover Institution.

governments and other power centers to an announcement of an ETI detection in ways that cannot now be anticipated.

Access to Information

Access to power is related to access to information. Roughly speaking, greater access to power and information allows individuals more potential for response and satisfaction whereas lower access leads to apathy and frustration. Greater access to accurate information is likely to give an individual a better grasp of the issues surrounding SETI. This does not guarantee, however, that that individual will necessarily accept SETI's premises or aims. Fraknoi's (1988) survey of science reporters, for example, seems to indicate that some sci-

ence reporters are not, in the absence of a verified signal, supporters of SETI, notwithstanding their full access to information and scientific understanding of the issues.

In relation to SETI, the greater the dissemination of accurate information about a signal detection and the more that individuals feel they can affect decisions about it, the more likely they are to view the event with interest. The less information available and the less the feeling of ability to affect subsequent decisions, the more apathetic an individual is likely to be about the event. Widespread sharing of information about SETI and its aims and procedures and promotion of the idea that all should participate in some ways in any decision about a reply would promote positive feelings about ETI detection and would tend to limit adverse public reactions. In this perspective, actions taken by SETI researchers to delegate decisions about a reply (and other decisions) to the UN or some other representative deliberative body would probably promote positive feelings about an ETI detection.

The Links Between the Variables

Access to power and information, regional identity, level of education, ethnicity, religion, gender – all of these affect peoples' responses to major events and announcements. We are presently able to broadly anticipate ways in which each of these individual attributes might influence a person's response to an ETI announcement, and it is likely that directed study could sharpen our focus considerably. More problematic is the fact that we have no present framework for anticipating how the attributes interact – say, how or whether a response by someone with great access to power is tempered by his or her regional affili-

ation or religion. Identifying ways in which the variables interact to modify behavior is of foremost importance, if behavioral sciences are to be applied to help envision public responses to an ETI announcement.

The Social Contexts within which People Respond

In addition to social attributes that affect individual behavior, external social contexts apply in more-or-less equal measure to all individuals in a given culture at a given time. Cultures emerge, persist, and change within the context of a specific environment and historical moment (Steward, 1963). Here we identify important social contexts that seem comparable across cultures and that appear to be influential in determining human responses to an ETI detection event.

The Emotional Context of Everyday Life

The “emotional context” of a culture underlies and motivates people’s ongoing interpretations of natural, social, and political events.

Emotional context may be measured on a continuum whose opposite poles we call *anticipation* and *anxiety*. We define “anticipation” as an emotional environment in which people have a generally positive outlook with respect to future events, are able to contemplate change with low levels of fear, and view negative events as isolated and exceptional. We define “anxiety” as an emotional environment in which people’s views of the future and possible changes are, on the whole, fearful and negative. In “anxious” times, negative events are interpreted as parts of a pattern that is generally perceived as deleterious. Events that do not conform to negative expectations are also interpreted as somehow threatening or nega-

tive (“They’re just trying to lull our suspicions before they *really* hit us”). The emotional context of the times strongly affects the ways in which people react to headline news (and other events) that they perceive as affecting their interests.

National global emotional contexts are engendered by present (and immediate past) experience of events. A populace in a world in which international tension is low, tolerance of differences is high, prosperity is generally on the rise, and perceived disjunctions in people’s lives are few, is likely to have positive expectations. On the other hand, if the world has been subject to a series of widespread natural disasters or political upheavals just before an ETI signal detection, such factors may provoke an anxious emotional climate and negatively affect the way in which the signal is perceived. If one culture (or sub-culture) is subject to positive or stressful events that are contrary to the national or worldwide context, the responses of that culture might differ significantly from those in other parts of the world.

Humanity’s global emotional context will probably be somewhere between the polar opposites we have chosen, with local populations at slightly different points on the scale. Truly extreme responses to an ETI signal seem relatively unlikely under “normal” conditions but would be more likely against a background of extremely stressful political or natural events. To clarify, if the world were to benefit from a decade of peace and prosperity, an extremely positive emotional context would be established. The reverse would be true in case of a decade of war and natural disasters. The extreme polar emotional contexts seem unlikely for the foreseeable future but a spread of less-extreme contexts can be anticipated for various nations, drawing upon and contributing to the average global context.

The Importance of Language

Language, particularly as it is used in a particular culture to name things and describe events, has an important (if sometimes unnoticed) effect on the way in which those things and events are perceived. Language (and image repertoires to be discussed in the next section) forms the cultural context within which members of a culture are able to envision, articulate, and imagine things other than themselves. Therefore it is important to identify the overtones of the linguistic terminology used in relation to SETI and the historical image repertoires that underlie popular imagination and description of extraterrestrial beings. The cultural variations and values of language and imagery and their "elite" and "popular" variations and values within any single culture require analysis (Redfield, 1956). Such analyses will be central to finding the appropriate terms and images needed to inform various publics about SETI as well as to prepare them for the possibility of an eventual signal detection.

For example we need to recognize crucial distinctions between the terms "signal" and "message." While the two are used as neutral technical terms by scientists and engineers, they have other connotations for the general public. For some, the terms are synonymous. The word "signal," as confused with "message" or even "sign," may well be endowed with greater emotional value than scientists commonly realize and may be taken as having potentially great moral, scientific, or other import by many people. SETI scientists are sensitive to this issue. For example, they usually avoid using the term "alien," because one of its alternate meanings is "repugnant". It is for this reason that "alien" is not used in this report. The terms "extraterrestrial" or "ETI" are used instead.

Awareness of language use is crucial in understanding other cultures with their different potentials for responses to an ETI signal detection and in translating information about SETI and an ETI signal from one culture to another. *Uchujin* is the word for "space person" in Japanese, while *chayzar* is the term for "stranger-life" in Hebrew. What are the values and connotations prompted by the use of these words in their respective cultures? Are there synonyms in each culture that have different connotations? Even in translation, "space-person" and "stranger-life" are not precisely synonymous. The connotations of words related to SETI as understood in their cultural contexts need analysis and additional research, and the use of synonyms with negative connotations should be avoided.

The Importance of Image Repertoires

An "image repertoire" is the store of images and dramatic scenarios that members of a culture or subculture share and draw upon (consciously and subconsciously) in trying to imagine the possibilities inherent in situations of which they have little or no experience (Sobchack, 1982; Sobchack and Sobchack, 1987). The repertoire includes images and scenes from cultural myths (Washington and the cherry tree), literature (the Southern belle), familiar artwork (Whistler's mother), popular imagery (the cowboy or cattle drive in the West), and media (the Huxtable family in a television situation comedy) and have no necessary relationship to scientific or factual data. The historically accumulated repertoire of images and dramatic scenarios available to a culture provides an important context for interpreting *new* images and events.

Understanding differences in the image repertoires of different cultures is important to

understanding how SETI and a detected signal might be specifically perceived. For example, in the past two decades, three major images of non-human intelligences have emerged in the US – all of them largely the result of mass media narratives such as films.

First is the image of the “colonizing alien” portrayed in *The Thing* and *War of the Worlds*: a menacing entity intent on bodily harm or physical, mental, and moral domination. The force of this image is traceable in part to the Cold War period, during which extraterrestrial invasions were linked in the public mind with Communist invasion scenarios. The image is also associated with the ambivalent public feelings about “advanced” technology of that time, which grew in extent after the advent of the atomic bomb.

A second image is that of the “benevolent alien,” the extraterrestrial of advanced intelligence who represents some higher moral force and would “save us from ourselves.” The benevolent alien warned against global conflict, possessed superior powers, and was usually more friendly, cuddly and “innocent” than its menacing counterpart. Friendly beings of this sort are portrayed in the films *Close Encounters Of The Third Kind* and *ET*. Finally there is a current prevalent image of the “cyborg”, a not-quite-human computerized intelligence envisioned as both menacing and protective, as a *Terminator* or *Robocop*. While linked in some ways to the Cold War image repertoire surrounding computers and robotics, the contemporary image plays to the pervasive public knowledge of microchips, biotechnology, and artificial intelligence, and a tendency to attribute intentionality and emotion to complex technological artifacts like the computer (Sobchack, 1987).

An important distinction in many cultures can be made between “popular” and “elite” image

repertoires. While both the intellectual elite and members of popular culture share many cultural images, their repertoires also differ. In considering evidence of an ETI detection, the elite in the US may, for instance, draw not only upon pervasive cultural images of extraterrestrials like Chewbacca or ET but also upon the extraterrestrials imagined by more philosophical science fiction writers (for example, Ursula LeGuin) or by more esoteric and philosophical movies than *Star Wars*. Many members of the general public in popular culture will draw not only upon shared images such as Chewbacca and ET but also upon the contributions of supermarket tabloids (“Martians Stole My Baby!”) Many of the popular images and scenarios are perpetuated regularly by tabloid journalism (among other media). Because catastrophe sells newspapers, there is no reason to expect that such media will feel any particular responsibility to “imagine” extraterrestrials as benign and unthreatening. Even if the tabloids were to take a millennial view, in all likelihood they would also appeal to more populist sentiments: “The Day Of Revelation Is Upon Us.”

Image repertoires may counteract rational explanations of events. Although scientists may explain that an ETI signal comes from 1000 light years away, and that that vast distance minimizes the possibility of interstellar travel and any immediate physical threat, a large portion of the public accepts the notion of “warp drive” and “warp speed” as presented by films and TV programs such as *Star Trek*. No matter what scientists say, there will always be some sense among the public that if an ETI culture has the technology to send a signal, then its members can be expected imminently in person, as well. This feeling emerges in some of the studies cited earlier (for example Ashkenazi, 1991).

The Importance of Belief Systems

As all societies have economic and political systems, so do they also have “belief systems” about reality, knowledge, value, the meaning of life, and other philosophical concepts. Belief systems generally fall into the category of religious belief, but are not confined to it. Political ideologies as well as secular moral codes are included too. Often belief systems are strongly grounded on religious viewpoints (“In God we trust”), but they transcend religion in many dimensions, dealing also with political and social views that are specific to a nation (such as “The American people”). Belief systems generally find their expression in symbols that both motivate action and suppress careful examination of specific issues. Such symbols can motivate both positive and negative action. The things they mean to those who hold them are generally extremely broad and complex (Turner, 1967).

In considering the influence of belief systems on reactions to SETI, we should consider three parameters: (1) how widespread is acceptance of the belief system, (2) how intensely is it held, and (3) what “permitted variations” (if any) are perceived by the adherents.

The Importance of Economic Motivation

All people live and work within economic systems that are based upon production, consumption, and exchange. (The term “economic” goes well beyond Western economics and the particular “positivistic” and profit-oriented motivation that is implied in the West.) There are both commonalities and differences among economic systems of different cultures. The economic variable is important to SETI because economic considerations will likely be affected by detection of an ETI signal.



Gary Coulter, HRMS Program Manager for NASA.

Reciprocally, the economic context in which a signal is detected would have an effect on the public response to it.

In addition to individual economic actors, modern societies are characterized by complex organizations that are dedicated largely to managing economic affairs. The integrated “economic community” of labor unions, banks, investment groups, stock markets and the like would certainly assess the potential of a detected ETI signal for hints of possible economic opportunity or threat. The more unambiguous the ETI signal, the more definite and interpretable the content of the message, and the more immediate the likelihood of two-way communication between ourselves and the ETI’s, the stronger would be the reaction. News of an advanced technological society at a two-way communication distance of only twenty light-years would probably cause a

major flurry in world stock markets. News of a weak information-poor signal from a distance of two thousand light years is less likely to create major economic disturbances because the issues of important new applicable information and a practical response are less immediate.

Analyzing Human Responses

The Continuum of Likely Responses

The preceding describes only a few of the factors (and relationships between factors) that influence human behavior, with emphasis on those that seem in our opinion to be particularly salient. We now consider peoples' potential responses to SETI events as a function of individual attributes, group attributes, and social contexts.

Human responses to significant events such as an ETI signal constitute a continuum, rather than a set of discrete behaviors. For heuristic purposes, however, it is helpful to formulate categories of response as though they were arranged in sets of paired opposites, centered on a neutral response of "indifference" to identification of an ETI signal. In the "positive" direction, responses to an announcement can range from indifference through mild curiosity and exuberance to "pronoia" – the latter an irrational highly-motivating belief that manifests itself in hyperactive involvement. "Negative" responses can range through mild resentful curiosity, defensiveness, and paranoia. The different sets of opposites are traceable to the salience of an event for those affected by it and to its immediacy for them, and categorize the extent of their reactions to it.

Indifference

We start with the midpoint in the spectrum of possible responses to a SETI signal, which we call "indifference." Some individuals and groups, even though possessed of information about an event and its possible outcomes, are likely to be "indifferent" in the sense that they have no interest, either positive or negative, and take no actions at all. (Such persons may include those who feel that they will not be affected by the discovery in any way.) This non-response serves as a starting point from which it is possible to examine more animated responses.

It is not clear what proportion of humanity would react with true indifference to news of the receipt of an ETI signal. Even people with little knowledge or no previous opinion on the subject are likely to have initial thoughts that are affected by the factors discussed previously. Whether they are informed about SETI or not, an announcement would probably be discussed. We anticipate that most people would then react in some manner to satisfy their initial curiosity. Given the prevalent belief in the existence of ETI's as evidenced by the Gallup poll results, and the prolonged low-level exposure to SETI research that seem likely in the future, fewer people seem likely to react with true "indifference" as time goes by.

Positive or Negative Curiosity: Acceptance and Rejection

People often respond to events with little more than curiosity when they are persuaded that the event will have little immediate impact (positive or negative) on their lives. Media saturation in the final decade of the twentieth century has created a populace of "news voyeurs": individuals who eagerly view news



Vera Buescher of the SETI Institute, the Workshop Coordinator.

events but whose subsequent reactions are usually minimal. Such people often exhibit curiosity about the events that they witness, but in terms of action they remain uninvolved. Negative and positive curiosity do not stimulate people to action, at least immediately. However, initial curiosity may serve as a basis for greater involvement later. It is possible that encouraging curiosity (preferably positive) about the SETI enterprise in advance of a signal detection would induce people to be more positively oriented toward the emergent issues if a signal were received and to react in what the SETI community would call "positive" ways.

The most common human response to new and unexpected information is to try to fit the

new data into existing frames of thought and belief. Theories of cognitive dissonance suggest that when new information does not fit within established patterns, the information will be rejected and rationalizations will be found for this rejection. When new information exhibits features that may be congruent with existing patterns, it is more readily integrated with them. For SETI, this means that there will be a large group that is predisposed (even before signal receipt) to accept the reality of an ETI signal and which will perhaps even view it as a natural or expected event.

As noted earlier in this chapter, certain pilot studies suggest that acceptance of the idea of ETI is quite common, particularly among individuals with higher levels of formal schooling. Rejection of the idea of ETI as real entities is quite high among members of groups whose cognitive or political position would be threatened by information about a signal (for example, fundamentalist church members; see Rosenbaum, Maier, and Lavrakas, 1979). Such individuals will likely seek reasons to declare the signal a fraud or the research a scientific boondoggle.

Exuberance or Defensiveness

Human responses to great events always include both "exuberant" and "defensive" expressions, depending upon the events and the postures of the affected individuals and groups. For many people, the detection of a signal would be a significant event and would be exciting and stimulating. For others, detection would be significant but it would be negatively perceived: fear, worry, xenophobia and other defensive attitudes and feelings would come to the fore.

"Exuberant" reactions would tend to prompt

active searches for additional information (through media, social networks, libraries or other sources) and perhaps further actions. Defensive reactions would also motivate action, but with different intent: to protect the threatened persons or their frames of reference and membership groups from expected dissonant effects of the event. Two elements of the exuberant-defensive category are important for SETI researchers to consider. They are, (1) the responders will search for more information, and (2) they will aggregate in groups of individuals with similar positions.

Individuals reacting exuberantly or defensively will want to know more about the phenomenon in order to bolster their positions. The sources that individuals ordinarily consult for information are expanded under such circumstances. (For example, if television is normally the major news resource, an individual in "exuberant" or "defensive" circumstances will expand his or her purview to include newspapers, journals, or other sources). Those who react exuberantly or defensively will also tend to aggregate in groups that are dedicated to mutual support of their positions and to appropriate actions. In extreme cases, such groups will be physically and vocally conspicuous in pressing their viewpoint.

A moderate case of exuberance or defensiveness would verge on the millennial or catastrophic responses described in Chapter 2. Such responses, based upon feelings that the detection of an ETI signal has overwhelming or even "ultimate" significance, can go in two diametrically opposite directions. Millennial responses would stem from a view that the signal event would not only upset the normal run of events but (even more) that it would bring about human "salvation" in some form. At the other extreme are apocalyptic responses based upon feelings that the signal event spells global or group catastrophe. Thus, for

example, the appearance of Halley's comet in the nineteenth century was hailed by some as a sign of coming salvation whereas others saw it as an indication that the end of the world was near.

Millennial and catastrophist responses to an ETI signal would likely occur among groups and individuals who have already had some intellectual (and perhaps organizational) predispositions toward these outcomes. UFO-ologists of various stripes, certain religious groups, and perhaps adherents of some academic schools of thought are likely to react in one of these fashions.

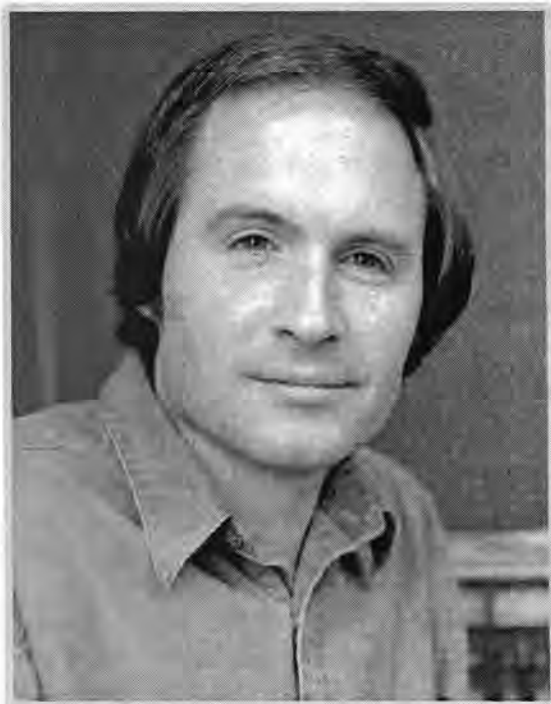
People who respond in this way are rarely critical of their own positions or of the data they absorb and employ. Millennial or catastrophist respondents would tend to accept some of the information about an intercepted ETI signal and reject other data that do not fit the scenario of salvation or doomsday that they envision. Millennial and catastrophist respondents also tend to react more vigorously than by mere discussion. If experience is any guide, they are likely to organize into large support groups and might exhibit somewhat strange (or even extreme) social and economic behaviors. These may include participation in rituals and convening to "await the aliens," as has been the case contemporaneously with New Guinea tribesmen who attempted to encourage the return of the US army by constructing "landing fields" in the brush, complete with wooden "airplanes" (see Chapter 2).

Irrationally Extreme Responses; Paranoia and Pronoia

The continuum of responses includes behaviors that can be even more extreme. An overwhelming and (in the specific terms of its culture) "irrationally" positive response might be

called “pronoia” (Finney, 1986). Its opposite is a similarly overwhelming irrationally negative response, paranoia. In both cases the reality of the signal or event is not disputed, but the person’s extreme valuation and interpretation of it cannot be shaken by argument, criticism, or contradictory data. In the event of an ETI signal, such responses are likely to be infrequent. Since the affected persons can be led to extravagant acts, however, they should be considered.

Pronoid responses that positively overvalue the significance of an ETI signal might take the form of deification of the signal originators. This response may or may not be related to responses by organized religious groups. A pronoid response might take the form of attempts to communicate with the signal originators, attempts to collect information by irregular means, issuance of public claims of



Bob Arnold, of the SETI Institute.

special knowledge, and perhaps harassment of SETI scientists and officials with demands for recognition.

Paranoid responses may resemble pronoid responses on occasion, even if their underlying motivations are overwhelming fear rather than compulsive attraction. Paranoids, like pronoids, may deify the signal originators and attribute great but terrifying and malevolent power to them. The paranoid response might also generate “conspiracy theories,” both about the ETI’s and SETI scientists, that are acted upon by some of the responders.

These responses are likely to be restricted to individuals (particularly among those who are clinically unbalanced), but they may manifest themselves as group actions, as well. The size of the population manifesting paranoid or pronoid behaviors will fluctuate as a function of the circumstances detailed previously in this chapter. Paranoid or pronoid individuals will react forcefully, perhaps violently, to news about an ETI signal. They will be persistent and indefatigable in attempting to acquire information about the signal and about its originators and (quite possibly) in insisting that their point of view not only be heard but that their plans of action be adopted to the exclusion of other possibilities.

Predicting Human Responses

Collective as well as individual responses to a SETI detection event will fall somewhere along the continuum that we have presented here. We cannot determine at this time which combinations of individual attributes in conjunction with any particular social contexts will cause an individual or group to respond in any particular mode. Lacking previous exposure to ETI signals, we cannot draw upon

analogous experience to envision how any particular group would respond to news of an ETI detection. For the pre-detection interval of the near future, however, we may begin with an approach like that of Donald Tarter (1991), who outlined ways in which variables in the signal might influence the relative numbers of individuals or groups responding in particular ways, the intensities of their responses, and possible actions that groups might undertake as parts of their responses.

Tarter has suggested that any suspected ETI signal we might receive can be examined for five signal characteristics: (1) certainty [of ETI origin]; (2) clarity [of signal content]; (3) "sensory accessibility" [from audible through visual to olfactory, perhaps tactile senses]; (4) activity level [demanded of the recipients, namely humanity]; (5) intentionality [signal overheard, or directed at us specifically] (D. Tarter 1991, pp. 6-16). Responses to each factor are increasingly modified by the character of the previous factors. During an ETI signal detection event, whether scientifically confirmed or only perceived as such, these signal characteristics would strongly influence human responses. For example, there might be many more groups and individuals responding in positive uncritical ways if the content of a signal were interpretable as promising great benefits to mankind – a "Galactic Encyclopedia" message – than if it were ambiguous or impossible to interpret. By anticipating signal features and various signal detection scenarios as Tarter has done, and by following up with surveys and polls on likely views and responses in those events, we may more closely delineate the ranges of likely responses and perhaps obtain a preview of some surprising outcomes.

To put human response categories in perspective, we suggest a graphic analog. Patterns of response can reasonably be expected to follow

a normal distribution. The more extreme responses will be fewer and the responses of most people will cluster in the middle. Under different contexts, this normal distribution will be perturbed. Under worldwide or national conditions of anticipatory emotional context, responses will be tilted in the positive direction (toward exuberance), with a tilt toward defensiveness more likely in an "anxious" emotional context.

The Time Dimension: Pre-event, Short-term, Long-term and Zero-event Responses

Responses to events related to detection and verification of an ETI signal can be divided into three phases: pre-detection anticipation, immediate responses at the time of the detection, and responses subsequent to detection.

During the pre-detection phase, anticipation of signal reception may gather force and may result in activities such as the formation of interest groups. There may be persistent and increasing interest in SETI as an unusual scientific endeavor. However, this phase is a period in which there will be little information with which most people would be able to frame their perspectives. The length of the pre-detection phase would probably have an inverse effect on interest and support for SETI. The longer the pre-detection period lasts, the less interest the search would engender. Given that a number of other factors might affect responses, as we noted above, the best we can say is that the response patterns will be indeterminate but will tend toward the middle of the response spectrum we have proposed. Most people will be indifferent, smaller numbers curious, a few well-organized groups (such as members of the HRMS proj-

ect) exuberant, and still others (perhaps members of Congress) suspicious about its outcome. Paranoid and pronoid reactions would be limited to a few individuals.

The state of public interest is likely to change abruptly after detection and verification of an ETI signal. At that time, responses will coalesce into the categories (or subvariants) that we have identified. These immediate responses seem to have the most potential for negative or positive consequences and are perhaps the most amenable to modification by advance preparation.

In the long run, responses to an ETI signal depend on many factors that cannot be detailed at present. Essentially the interaction of individual social attributes and social contexts with the signal typology and the time needed for confirmation and interpretation of the signal will determine which portions of a given population exhibit the various response types. Long term responses may undergo continual flux as new information becomes available and as global emotional contexts change.

The possibility of “no detection” should be considered as carefully as the possibility of a detection. No one in the SETI community anticipates that a signal will be detected soon after the major search begins. The signal detection process might continue for decades or perhaps even for centuries without revealing evidence of ETI’s. Disregarding the effect of search duration on specific SETI projects (such as NASA’s), the length of the search does not change the method of analysis. We may assume that education and information on SETI will become more readily available worldwide (because that is the current, though not inevitable, trend) and therefore we may also assume that – barring major changes in social contexts – responses to SETI will proceed much as before, with most of the non-



Amahl Drake, Workshop Co-coordinator, helped keep the proceedings on track.

informed population indifferent, with others more or less curious, with some small groups exuberant or defensive, and with fewer holding extremist views.

Recommendations

Further research will modify and perhaps alter the methodological approach that we have suggested in this chapter, and our tentative conclusions. Many of the relationships between factors mentioned here that are relevant to the behavioral sciences’ approach to SETI are presently unknown, but they are knowable. Our recommendations identify work needed to acquire basic data. Additional programmatic suggestions are also presented in a separate section.

Our foremost recommendation is for further detailed research on the issues we have outlined above. Such research should include a thorough examination of:

- Literature and published reports on mass psychology;
- Individual and public responses to cultural differences;
- Individual and public reactions to perceived “strangeness” and to science and technology;
- The shades of meaning of linguistic terms used in connection with SETI and in media presentations of ETI.

We also encourage research into specific topics relating to SETI, such as cross-cultural studies of popular images of ETI. Detailed suggestions follow.

1. Many cultures have traditions of depicting strangers, aliens, and ETI, and have different emotional responses to specific terms used within those traditions. Analyzing these terms (and perhaps coining new ones) will enable SETI participants to anticipate and pre-empt possible negative responses by adopting accurate language that is free of unexpected shadings of value.
2. There is need for research on cross-cultural image repertoires of ETI’s and on cultural conceptions of science and technology. Some such studies have been carried out, but better summaries of findings and cross-cultural comparisons of the findings are still lacking.
3. We need to develop better understanding of the links between the variables in our

methodological scheme. The preliminary framework that we have adopted here identifies the variables – elements of social context and personal and group attributes – but does not specify the ways in which the elements interact to influence behavior. It is important to consider how these various elements fit together.

4. There is a need to identify groups with unusual influences whose members might require better information about SETI. What are the interests of the members, and can better understanding of SETI encourage them to view these endeavors positively (or more positively)?
5. We should try to identify and clarify the responses of various peoples to the Apollo and Viking missions (activities that were remotely analogous to NASA’s HRMS research).
6. SETI researchers (or a related organization) should identify individual and institutional responses that are viewed by the public as preferred and achievable in the event of a signal detection, then participate in programs of education and information that will enable people to respond in those ways. Do not assume a one-to-one correlation between education about SETI and positive responses to SETI concerns and discoveries.
7. SETI researchers or a related organization should make greater use of popular media, including movies, computer games, and popular music, to present SETI and ETI themes. The few studies available have revealed that people are interested in SETI and that there seems to be a correlation between exposure to scientific experience and positive views of the existence of ETI. Popular media, computer games, and pop-

- ular music could increase such interest and exposure.
8. SETI researchers should assist national and international organizations concerned with SETI-related issues. Such organizations often have subtle (or sometimes overt) positive effects on their associates. They tend to be enthusiastic, are inclined to positive action, and foster some of the attitudes in which SETI researchers are interested, if only by ripple effect. Such groups might include astronomy clubs, high school science fair organizations, and ham radio associations.
 9. Firm relationships with professional and intercultural academic/popular bodies should be encouraged so as to draw upon multiple disciplinary resources for understanding those variables that are unclear regarding human reactions. Thus, SETI researchers should associate with "soft" sciences through such bodies as the World Ethnographic Union, in a form similar perhaps to Bioastronomy Commission 51 of the IAU. Multinational organizations such as the United Nations should be informed of the potential importance of SETI for humanity as a whole.
 10. The creation of a panel of expert behavioral scientists as a "reaction team" should be considered. Such a team should be available for advice and help in cases where information about an ETI signal seems to cause unusual disruption of normal patterns of life. Instances of mass hysteria that have sometimes accompanied unusual events (for example, the "cattle mutilation epidemic" in Nebraska in 1974, blamed on visiting ETI's, or instances of pronoid reactions to a real ETI message could be addressed by such a team.
 11. Most members of the SETI project (and of this present panel) are White Americans. It is suggested that means be devised for including the perspectives and inputs of members of non-American, non-White societies in the future. Future research must take into account the world emotional context and perspectives of other cultures.
 12. Great care must be taken that methodological assumptions are not made into metaphysical assumptions in describing SETI activities. For example, we should not assume as a matter of dogma that ETI civilizations are eager to communicate with us, nor that all human beings are eager to communicate with them.